

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 12A.3

UNDERSTANDING ENTROPY CHANGES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	12: Entropy and Energetics
SUB-TOPIC:	12A.3: Understanding Entropy Changes
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 12A.3

Q1. Explain, in terms of entropy, why a reaction is feasible at all temperatures if ΔH is negative and ΔS_{system} is positive. [WCH14/01/Oct21/Q4(b)]

[3 Marks] **WCH14/01**

Q2. Calculate the minimum temperature at which a reaction becomes feasible given $\Delta H = +135 \text{ kJ mol}^{-1}$ and $\Delta S_{\text{system}} = +334 \text{ J K}^{-1} \text{ mol}^{-1}$. [WCH14/01/Jan21/Q6(c)]

[2 Marks] **WCH14/01**

SECTION B: WORKED MARK SCHEME — 12A.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARK S
Q1	Q1: Explain, in terms of entropy, why a reaction is feasible at all temperatures if ΔH is negative and ΔS_{system} is positive. [WCH14/01/Oct21/Q4(b)] - $\Delta S_{\text{surroundings}} = -\Delta H / T$, so if ΔH is negative, $\Delta S_{\text{surroundings}}$ is positive at all T (1) - $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$, so ΔS_{total} must be positive (1) - Since $\Delta S_{\text{total}} > 0$, the reaction is feasible (1)	[3]
Q2	Q2: Calculate the minimum temperature at which a reaction becomes feasible given $\Delta H = +135 \text{ kJ mol}^{-1}$ and $\Delta S_{\text{system}} = +334 \text{ J K}^{-1} \text{ mol}^{-1}$. [WCH14/01/Jan21/Q6(c)] For feasibility, $\Delta S_{\text{total}} = 0$ (minimum condition) So $T = \Delta H / \Delta S_{\text{system}}$ (1) = $(+135000 \text{ J mol}^{-1}) / (+334 \text{ J K}^{-1} \text{ mol}^{-1}) = 404 \text{ K}$ (1)	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 12A.3**FAQ #1: Why do students struggle with finding the minimum temperature for feasibility?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why do students struggle with finding the minimum temperature for feasibility?

MODEL ANSWER / EXAMINER EXPECTATION:

Candidates often set ΔS_{total} to 0 but fail to convert ΔH to J mol^{-1} . This gives a completely wrong temperature. Always ensure ΔH and ΔS_{system} have the same energy units before dividing.

FAQ #2: What is meant by thermodynamic feasibility?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is meant by thermodynamic feasibility?

MODEL ANSWER / EXAMINER EXPECTATION:

Thermodynamic feasibility means ΔS_{total} is positive. It indicates a reaction can occur, but it does NOT mean the reaction will happen quickly (kinetic aspect is ignored).